

Existence and Uniqueness for Low-Rank Mean-Field Neural Networks

1 Existence and Uniqueness of the Solution of the Mean-Field ODEs

This section establishes well-posedness of the mean-field ODE system (??)–(??) for low-rank networks with L layers and r channels. The proof adapts the Picard iteration argument from [?] to account for the low-rank structure, showing that the solution operator is contractive on appropriate function spaces.

1.1 Setup and Notation

We consider the mean-field ODE system:

$$\partial_t w_1(t, c_1, k) = -\xi_1(t) \mathbb{E}_Z \left[d_L(Z; W(t)) \varphi_1(\langle f_1(c_1), X \rangle) B_k^{(2)}(t; X) \right], \quad (1)$$

$$\partial_t w_i(t, c_i, k) = -\xi_i(t) \mathbb{E}_Z \left[d_L(Z; W(t)) \varphi_i(H_i(t, c_i; X, W(t))) B_k^{(i+1)}(t; X) \right], \quad i = 2, \dots, L-1, \quad (2)$$

$$\partial_t w_L(t, c_L) = -\xi_L(t) \mathbb{E}_Z \left[d_L(Z; W(t)) \varphi_L(H_L(t, c_L; X, W(t))) \right], \quad (3)$$

where for $i = 2, \dots, L-1$:

$$B_k^{(i)}(t; X) = \mathbb{E}_{C_i} \left[L_{C_i, k}^{(i)} \varphi'_i(H_i(t, C_i; X, W(t))) w_i(t, C_i, k) \right],$$

and $H_i(t, c_i; X, W) = \sum_{k=1}^r L_{c_i, k}^{(i)} m_k^{(i)}(t; X, W)$ with $m_k^{(i)}$ defined recursively.

1.2 Norms and Function Spaces

We equip the mean-field parameters with several norms to control growth and establish contractivity. For the low-rank case with L layers and r channels, we define:

Orlicz-type norms:

$$\llbracket w \rrbracket_1 \llbracket \cdot \rrbracket t = \max_{1 \leq k \leq r} \mathbb{E}_{C_1} \left[\sup_{s \leq t} |w_1(s, C_1, k)|^{50} \right]^{1/50}, \quad (4)$$

$$\llbracket w \rrbracket_i \llbracket \cdot \rrbracket t = \max_{1 \leq k \leq r} \mathbb{E}_{C_i} \left[\sup_{s \leq t} |w_i(s, C_i, k)|^{50} \right]^{1/50}, \quad i = 2, \dots, L-1, \quad (5)$$

$$\llbracket w \rrbracket_L \llbracket \cdot \rrbracket t = \mathbb{E}_{C_L} \left[\sup_{s \leq t} |w_L(s, C_L)|^{50} \right]^{1/50}, \quad (6)$$

$$\llbracket W \rrbracket \llbracket \cdot \rrbracket t = \max(\llbracket w \rrbracket_1 \llbracket \cdot \rrbracket t, \max_{2 \leq i \leq L-1} \llbracket w \rrbracket_i \llbracket \cdot \rrbracket t, \llbracket w \rrbracket_L \llbracket \cdot \rrbracket t). \quad (7)$$

L^2 -type norms:

$$\|w_1\|_t = \max_{1 \leq k \leq r} \mathbb{E}_{C_1} \left[\sup_{s \leq t} |w_1(s, C_1, k)|^2 \right]^{1/2}, \quad (8)$$

$$\|w_i\|_t = \max_{1 \leq k \leq r} \mathbb{E}_{C_i} \left[\sup_{s \leq t} |w_i(s, C_i, k)|^2 \right]^{1/2}, \quad i = 2, \dots, L-1, \quad (9)$$

$$\|w_L\|_t = \mathbb{E}_{C_L} \left[\sup_{s \leq t} |w_L(s, C_L)|^2 \right]^{1/2}, \quad (10)$$

$$\|W\|_t = \max \left(\|w_1\|_t, \max_{2 \leq i \leq L-1} \|w_i\|_t, \|w_L\|_t \right). \quad (11)$$

Sub-Gaussian (ψ_2) norms:

$$\llbracket w_1 \rrbracket_{\psi, t} = \sqrt{50} \sup_{m \geq 1} \frac{1}{\sqrt{m}} \max_{1 \leq k \leq r} \mathbb{E}_{C_1} \left[\sup_{s \leq t} |w_1(s, C_1, k)|^m \right]^{1/m}, \quad (12)$$

$$\llbracket w_i \rrbracket_{\psi, t} = \sqrt{50} \sup_{m \geq 1} \frac{1}{\sqrt{m}} \max_{1 \leq k \leq r} \mathbb{E}_{C_i} \left[\sup_{s \leq t} |w_i(s, C_i, k)|^m \right]^{1/m}, \quad i = 2, \dots, L-1, \quad (13)$$

$$\llbracket w_L \rrbracket_{\psi, t} = \sqrt{50} \sup_{m \geq 1} \frac{1}{\sqrt{m}} \mathbb{E}_{C_L} \left[\sup_{s \leq t} |w_L(s, C_L)|^m \right]^{1/m}, \quad (14)$$

$$\llbracket W \rrbracket_{\psi, t} = \max \left(\llbracket w_1 \rrbracket_{\psi, t}, \max_{2 \leq i \leq L-1} \llbracket w_i \rrbracket_{\psi, t}, \llbracket w_L \rrbracket_{\psi, t} \right). \quad (15)$$

The factor $\sqrt{50}$ ensures that $\llbracket W \rrbracket_{\psi, t} \geq \llbracket W \rrbracket_t$ and hence $\llbracket W \rrbracket_{\psi, t} \geq \|W\|_t$. For convenience, we define the random variables:

$$\max_t^w(W) = \max_{1 \leq k \leq r} \sup_{s \leq t} |w_1(s, C_1, k)|, \quad (16)$$

$$\max_t^{w_i}(W) = \max_{1 \leq k \leq r} \sup_{s \leq t} |w_i(s, C_i, k)|, \quad i = 2, \dots, L-1, \quad (17)$$

$$\max_t^{w_L}(W) = \sup_{s \leq t} |w_L(s, C_L)|. \quad (18)$$

For two sets of mean-field parameters W and W' , we define the distance:

$$\|W - W'\|_t = \max \left(\|w_1 - w'_1\|_t, \max_{2 \leq i \leq L-1} \|w_i - w'_i\|_t, \|w_L - w'_L\|_t \right), \quad (19)$$

$$\|w_1 - w'_1\|_t = \max_{1 \leq k \leq r} \mathbb{E}_{C_1} \left[\sup_{s \leq t} |w_1(s, C_1, k) - w'_1(s, C_1, k)|^2 \right]^{1/2},$$

$$\|w_i - w'_i\|_t = \max_{1 \leq k \leq r} \mathbb{E}_{C_i} \left[\sup_{s \leq t} |w_i(s, C_i, k) - w'_i(s, C_i, k)|^2 \right]^{1/2}, \quad i = 2, \dots, L-1,$$

$$\|w_L - w'_L\|_t = \mathbb{E}_{C_L} \left[\sup_{s \leq t} |w_L(s, C_L) - w'_L(s, C_L)|^2 \right]^{1/2}.$$

1.3 A Priori Bounds

We first establish that any solution, if it exists, must remain bounded. This is crucial for defining the function spaces on which the solution operator acts.

Lemma 1 (A priori bounds). *Under Assumptions ?? and ??, given an initialization $W(0)$ with $\llbracket W \rrbracket_{\psi,0} < \infty$, a solution W to the mean-field ODEs, if it exists, must satisfy that for any $t \in [0, \infty)$:*

$$\llbracket W \rrbracket t \leq \llbracket W \rrbracket_{\psi,t} \leq K_0(t),$$

where $K_0(t)$ is a non-decreasing function of the form

$$K_0(t) = K^\kappa (1 + t^\kappa) (1 + \llbracket W \rrbracket_{\psi,0}^\kappa),$$

for some constant $\kappa > 0$ depending on K , r , and L . Furthermore, for any $B \geq 0$:

$$\mathbb{P}\left(\max\left(\max_t^w(W), \max_{2 \leq i \leq L-1} \max_t^{w_i}(W), \max_t^{w_L}(W)\right) \geq K_0(t)B\right) \leq Ce^{-K_1 B^2},$$

for some universal constants $C, K_1 > 0$.

Proof. The proof uses Grönwall-type arguments applied to the ODEs (1)–(3). The key steps are:

1. **Boundedness of drifts:** By Assumption ??:

- $|d_L(Z; W)| \leq K$ (bounded loss derivative)
- $|\varphi_i|, |\varphi'_i| \leq K$ (bounded activations)
- $|H_i| \leq \|L^{(i)}\|_{\infty,1} \max_k |m_k^{(i)}| \leq rK \max_k \mathbb{E}_{C_{i-1}}[|w_{i-1}(C_{i-1}, k)|]$ for $i = 2, \dots, L$
- $|B_k^{(i)}| \leq K \mathbb{E}_{C_i}[|L_{C_i,k}^{(i)}| |w_i|] \leq rK^2 \mathbb{E}_{C_i}[|w_i|]$ for $i = 2, \dots, L-1$

2. **Control of H_i with low-rank structure:** For $i = 2, \dots, L$:

$$\begin{aligned} |H_2| &\leq \|L^{(2)}\|_{\infty,1} \max_{1 \leq k \leq r} |m_k^{(2)}| \leq rK \max_{1 \leq k \leq r} \mathbb{E}_{C_1}[|w_1(C_1, k)|], \\ |H_i| &\leq \|L^{(i)}\|_{\infty,1} \max_{1 \leq k \leq r} |m_k^{(i)}| \leq rK \max_{1 \leq k \leq r} \mathbb{E}_{C_{i-1}}[|w_{i-1}(C_{i-1}, k)|], \quad i = 3, \dots, L. \end{aligned}$$

3. **Control of $B_k^{(i)}$ with low-rank structure:** For $i = 2, \dots, L-1$:

$$\begin{aligned} |B_k^{(2)}| &\leq K \mathbb{E}_{C_2}[|L_{C_2,k}^{(2)}| |w_2|] \leq rK^2 \mathbb{E}_{C_2}[|w_2|], \\ |B_k^{(i)}| &\leq K \mathbb{E}_{C_i}[|L_{C_i,k}^{(i)}| |w_i|] \leq rK^2 \mathbb{E}_{C_i}[|w_i|], \quad i = 3, \dots, L-1. \end{aligned}$$

4. **Grönwall argument:** From the ODE structure, we have for each layer:

$$|w_1(t, c_1, k)| \leq |w_1(0, c_1, k)| + \int_0^t K(1 + \max_{1 \leq k' \leq r} \mathbb{E}_{C_1}[|w_1(s, C_1, k')|] + rK^2 \max_{1 \leq k' \leq r} \mathbb{E}_{C_2}[|w_2(s, C_2)|]) ds,$$

and similarly for other layers. Taking expectations and applying Minkowski's inequality, then using Grönwall's lemma, we obtain polynomial growth in t . The key is that each layer contributes a factor $(1 + rK)$ to the Gronwall constant, giving $(1 + rK)^{L-2}$ total, which is polynomial in L (not exponential) because rK is fixed.

5. **Sub-Gaussian tail bound:** The tail bound follows from the ψ_2 control and a union bound over r channels and L layers.

The constant κ depends on K , r , and L through the factors $\|L^{(i)}\|_{\infty,1} \leq rK$ for $i = 2, \dots, L-1$, but remains polynomial rather than exponential in r and L because the factors multiply (not compound exponentially). $\square$

These a priori bounds lead us to consider the following spaces, given an initialization $W(0)$ and an arbitrary terminal time $T > 0$:

- The space $\mathcal{W}_T$ of mean-field parameters $W' = \{W'(t)\}_{t \leq T}$ such that $\llbracket W' \rrbracket T \leq K_0(T)$.
- The space $\mathcal{W}_T^0 \subset \mathcal{W}_T$ of mean-field parameters $W' \in \mathcal{W}_T$ such that:

$$\llbracket W' \rrbracket_{\psi, T} \leq K_0(T), \quad (20)$$

$$\mathbb{P}\left(\max\left(\max_T^w(W'), \max_{2 \leq i \leq L-1} \max_T^{w_i}(W'), \max_T^{w_L}(W')\right) \geq K_0(T)B\right) \leq Ce^{-K_1 B^2} \quad \forall B \geq 0, \quad (21)$$

and $W'(0) = W(0)$ (so all elements in $\mathcal{W}_T^0$ share the same initialization).

We equip these spaces with the metric $(W', W'') \mapsto \|W' - W''\|_T$. By Lemma 1, any solution W to the mean-field ODEs, if it exists, must belong to $\mathcal{W}_T^0$.

1.4 Solution Operator

Denote by $\mathfrak{W}_T$ the space of mean-field parameters W such that $\|W\|_T < \infty$. Given a terminal time $T \geq 0$ and an initialization $W(0)$, we define the mapping F that associates $W' \in \mathfrak{W}_T$ with:

$$F(W')(t) = \{F_1(W')(t, \cdot, \cdot), F_2(W')(t, \cdot, \cdot), \dots, F_{L-1}(W')(t, \cdot, \cdot), F_L(W')(t, \cdot)\}, \quad (22)$$

where for the first layer ($k = 1, \dots, r$):

$$F_1(W')(t, c_1, k) = w_1(0, c_1, k) - \int_0^t \xi_1(s) \mathbb{E}_Z \left[d_L(Z; W'(s)) \varphi_1(\langle f_1(c_1), X \rangle) B_k^{(2)}(s; X, W'(s)) \right] ds, \quad (23)$$

for intermediate layers ($i = 2, \dots, L-1$ and $k = 1, \dots, r$):

$$F_i(W')(t, c_i, k) = w_i(0, c_i, k) - \int_0^t \xi_i(s) \mathbb{E}_Z \left[d_L(Z; W'(s)) \varphi_i(H_i(s, c_i; X, W'(s))) B_k^{(i+1)}(s; X, W'(s)) \right] ds, \quad (24)$$

and for the output layer:

$$F_L(W')(t, c_L) = w_L(0, c_L) - \int_0^t \xi_L(s) \mathbb{E}_Z \left[d_L(Z; W'(s)) \varphi_L(H_L(s, c_L; X, W'(s))) \right] ds, \quad (25)$$

where $B_k^{(i)}(s; X, W')$ and $H_i(s, c_i; X, W')$ are computed with respect to W' .

Observe that at initialization $F(W')(0, \cdot, \cdot) = W(0)$, whereas the quantities in the time integrals are computed with respect to W' . A solution to the mean-field ODEs on $[0, T]$ is an element of $\mathfrak{W}_T$ satisfying $F(W) = W$. We say that W is a solution on $[0, \infty)$ if its restriction to $[0, T]$ is a solution on $[0, T]$ for all $T > 0$.

Lemma 2 (Solution operator maps bounded sets to bounded sets). *Under Assumptions ?? and ??, for any $W' \in \mathcal{W}_T^0$, we have $F(W') \in \mathcal{W}_T^0$.*

Proof. The proof follows the same argument as Lemma 1, using the integral form defining F (23)–(25) and the bounded/Lipschitz properties of the drifts. The key is that applying bounded/Lipschitz drifts to W' preserves the moment and tail bounds with constants controlled by $K_0(T)$. $\square$

1.5 Contractivity of the Solution Operator

The key technical result is that the solution operator is contractive on $\mathcal{W}_T^0$, which enables the Picard iteration argument.

Lemma 3 (Solution operator is contractive). *For a given $B \geq 0$, consider two collections of mean-field parameters $W', W'' \in \mathcal{W}_T$ such that:*

$$\mathbb{P}\left(\max\left(\max_T^w(W'), \max_{2 \leq i \leq L-1} \max_T^{w_i}(W'), \max_T^{w_L}(W')\right) \geq K_0(T)B\right) \leq Ce^{-K_1 B^2}, \quad (26)$$

$$\mathbb{P}\left(\max\left(\max_T^w(W''), \max_{2 \leq i \leq L-1} \max_T^{w_i}(W''), \max_T^{w_L}(W'')\right) \geq K_0(T)B\right) \leq Ce^{-K_1 B^2}. \quad (27)$$

Under Assumptions ??-??, for any $t \leq T$:

$$\|F(W') - F(W'')\|_t \leq (KK_0(T))^4(1+rK)^{L-2} \int_0^t \left((1+B)\|W' - W''\|_s + \sqrt{2}e^{-K_1 B^2/2}\right) ds,$$

where the constant depends on K , r , and L through $\|L^{(i)}\|_{\infty,1} \leq rK$ for $i = 2, \dots, L-1$.

Proof. The proof uses a good/bad event decomposition:

1. **Good event:** On the event $\{\max(\max_T^w, \max_{2 \leq i \leq L-1} \max_T^{w_i}, \max_T^{w_L}) \leq K_0(T)B\}$, the drifts are Lipschitz in W with constant $(1+B)K_0(T)$.

2. **Control of H_i differences:** For H_2 :

$$|H_2(W') - H_2(W'')| \leq \|L^{(2)}\|_{\infty,1} \max_{1 \leq k \leq r} \mathbb{E}_{C_1} [|w'_1(C_1, k) - w''_1(C_1, k)|] \leq rK \max_{1 \leq k \leq r} \|w'_1 - w''_1\|_t.$$

For H_i with $i = 3, \dots, L$:

$$|H_i(W') - H_i(W'')| \leq \|L^{(i)}\|_{\infty,1} \max_{1 \leq k \leq r} \mathbb{E}_{C_{i-1}} [|w'_{i-1}(C_{i-1}, k) - w''_{i-1}(C_{i-1}, k)|] \leq rK \max_{1 \leq k \leq r} \|w'_{i-1} - w''_{i-1}\|_t.$$

3. **Control of $B_k^{(i)}$ differences:** For $B_k^{(2)}$:

$$\begin{aligned} |B_k^{(2)}(W') - B_k^{(2)}(W'')| &\leq K \mathbb{E}_{C_2} [|L_{C_2,k}^{(2)}| (|w'_2(C_2) - w''_2(C_2)| + |w''_2(C_2)| |H_2(W') - H_2(W'')|)] \\ &\leq rK^2(1+B)\|W' - W''\|_t. \end{aligned}$$

For $B_k^{(i)}$ with $i = 3, \dots, L-1$, the difference is controlled similarly, giving a factor $(1+B)$ and accumulating factors of rK from each layer.

4. **Control of weight differences:** For the first layer:

$$\begin{aligned} |F_1(W')(t, c_1, k) - F_1(W'')(t, c_1, k)| &\leq \int_0^t K(1+B)K_0(T)\|W' - W''\|_s ds \\ &\quad + \int_0^t KrK^2(1+B)\|W' - W''\|_s ds \\ &\leq (KK_0(T))^2(1+rK)(1+B) \int_0^t \|W' - W''\|_s ds. \end{aligned}$$

For intermediate layers $i = 2, \dots, L-1$, each layer contributes a factor $(1+rK)$, giving $(1+rK)^{L-2}$ total. For the output layer, the bound is similar but without the B_k term.

5. **Bad event:** The bad event contributes an exponentially small remainder $\sqrt{2}e^{-K_1 B^2/2}$.
6. **Combining bounds:** Integrating in time and using Minkowski's inequality yields the result. The factor $(1+rK)^{L-2}$ comes from the accumulation of rK factors from each of the $L-2$ intermediate layers.

□

1.6 Main Theorem: Existence and Uniqueness

Theorem 1 (Existence and uniqueness of mean-field ODE solution). *Assume that the initialization $W(0)$ of the mean-field ODEs satisfies $\llbracket W \rrbracket_{\psi,0} \leq K$. Then under Assumptions ?? and ??, there exists a unique solution to the mean-field ODEs (1)–(3) on $t \in [0, \infty)$.*

Proof. We perform a Picard-type iteration argument. Consider an arbitrary finite $T \geq 0$ and $W', W'' \in \mathcal{W}_T^0$. From Lemma 3:

$$\begin{aligned} \|F(W') - F(W'')\|_t &\leq (KK_0(T))^4(1+rK)^{L-2} \left((1+B) \int_0^t \|W' - W''\|_s ds + T\sqrt{2}e^{-K_1 B^2/2} \right) \\ &\equiv k_1(1+B) \int_0^t \|W' - W''\|_s ds + k_2 e^{-k_3 B^2}, \end{aligned} \quad (28)$$

for any $B > 0$, where $k_1 = (KK_0(T))^4(1+rK)^{L-2}$, $k_2 = (KK_0(T))^4(1+rK)^{L-2}T\sqrt{2}$, and $k_3 = K_1/2$.

By Lemma 2, F maps $\mathcal{W}_T^0$ to $\mathcal{W}_T^0$. We can iterate the inequality (28) to obtain:

$$\begin{aligned} \|F^{(m)}(W') - F^{(m)}(W'')\|_T &\leq k_1(1+B) \int_0^T \|F^{(m-1)}(W') - F^{(m-1)}(W'')\|_{T_2} dT_2 + k_2 e^{-k_3 B^2} \\ &\leq k_1^2(1+B)^2 \int_0^T \int_0^{T_2} \|F^{(m-2)}(W') - F^{(m-2)}(W'')\|_{T_3} dT_3 dT_2 + k_2 \sum_{\ell=1}^2 \frac{(Tk_1(1+B))^{\ell-1}}{\ell!} \\ &\quad \dots \\ &\leq \frac{1}{m!} T^m k_1^m (1+B)^m \|W' - W''\|_T + k_2 e^{Tk_1(1+B) - k_3 B^2} \\ &\leq \frac{1}{m!} T^m k_1^m (1+\sqrt{m})^m \|W' - W''\|_T + k_2 e^{Tk_1(1+\sqrt{m}) - k_3 m}, \end{aligned} \quad (29)$$

where we choose $B = \sqrt{m}$ in the last display. Note that since $\llbracket W \rrbracket_{\psi,0} < \infty$, $K_0(T)$ and hence k_1, k_2 are finite for finite T .

By substituting $W'' = F(W')$, we obtain:

$$\sum_{m=1}^{\infty} \|F^{(m+1)}(W') - F^{(m)}(W')\|_T = \sum_{m=1}^{\infty} \|F^{(m)}(W'') - F^{(m)}(W')\|_T < \infty.$$

Hence as $m \rightarrow \infty$, $F^{(m)}(W')$ converges in $\|\cdot\|_T$ to a limit $W \in \mathfrak{W}_T$, which is a fixed point of F . By Lemma 1, W belongs to $\mathcal{W}_T^0$.

The uniqueness of the fixed point comes from the estimate (29), since if W' and W'' are fixed points of F , then they are both in $\mathcal{W}_T^0$, and:

$$\|W' - W''\|_T = \|F^{(m)}(W') - F^{(m)}(W'')\|_T \leq \frac{1}{m!} T^m k_1^m (1+\sqrt{m})^m \|W' - W''\|_T + k_2 e^{Tk_1(1+\sqrt{m}) - k_3 m},$$

and one can take m arbitrarily large. This proves that the solution exists and is unique on $t \in [0, T]$. Since T is arbitrary, we have existence and uniqueness of the solution to the mean-field ODEs on the time interval $[0, \infty)$. $\square$

1.7 Key Differences from Full-Width Case

The proof structure follows the full-width case [?], but with the following key adaptations for the low-rank structure:

1. **Norms account for r channels:** All norms take the maximum over $k \in \{1, \dots, r\}$ for intermediate layers, reflecting the r independent channels.
2. **Low-rank structure in bounds:** The pre-activations H_i involve sums over r channels with mixing matrices $L^{(i)}$, introducing factors of $\|L^{(i)}\|_{\infty,1} \leq rK$ in all bounds.
3. **Polynomial growth in L :** The Gronwall constant accumulates as $(1 + rK)^{L-2}$ (polynomial in L), not exponentially, because:
 - Each layer contributes a bounded factor $(1 + rK)$
 - These factors multiply (not compound exponentially)
 - All errors are bounded in terms of the global distance $\|W' - W''\|_t$, not layer-wise errors that would compound
4. **Frozen first layer simplifies base case:** Since $f_1(C_1)$ is frozen, $H_1(t, c_1; X, W) = \langle f_1(c_1), X \rangle$ is independent of $W(t)$, eliminating recursion at the base case and ensuring the pattern starts cleanly.

These adaptations ensure that the proof works for arbitrary depth L with polynomial (not exponential) growth in the constants, making the low-rank framework theoretically tractable for deep networks.